Name- Shivam Kumar
Section- 2B
Rollno.- 2415001503

Tree

Problem 1 (a) Leaf Node

Dev1, Dev2, HR, CFO

(b) Height of the tree.

Longest path:

CEO → CTO → Dev lead → Dev1

No of edges = 3

Height = 3

(c) Depth of "Dev Lead"

Path.

CEO → CTO → Deve Lead.

(d) Ancestors of "Dev1"

Dev Lead, CTO, CEO.

(e) Degree of the CTO Node

no. of children.

Degree of CTO = 2.

Problem 2:-

(a) Listing files in alphabetical order (BST case).

Use Inorder Traversal (Left → Root → Right).

(b) Calculating total directory size.

Visit all children first, then combine their sizes into the parent.

Use Postorder Traversal (Left → Right → Root)

ⓒ Creating a full backup

Copy the root first, then recursively copy children.

Use Preorder Traversal (Root - Left - Right).

ⓓ Traversals of the given tree

Inorder (Left → Root → Right)

config → user → home → docs → root → var - log.

Preorder (Ro L R)

root → home → user → config → docs → var → log.

Postorder (L R Ro).

Config → user → docs → home → log → var → root.

ⓔ Why Postorder is used for deletion

when deleting directories-

- You must delete all files/subfolders first.
- Then delete the parent directory.

Postorder ensures -

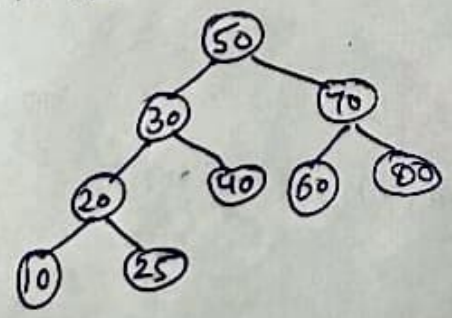
1. Children are deleted first
2. Parent is ~~las~~ deleted last.

Problem 3 -

ⓐ Resulting BST after Insertion

Insert in order.

50 → 30 → 70 → 20 → 40 → 60 → 80 → 10 → 25.



(b) Comparisons when searching for 25

- Compare with 50 → 25 < 50 → go left
- Compare with 30 → 25 < 30 → go left
- Compare with 20 → 25 > 20 → go right
- Compare with 25 → found.

Sequence → 50 → 30 → 20 → 25

(c) Inorder traversal

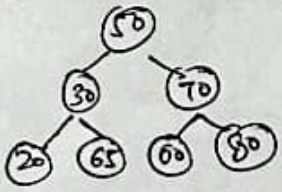
Left → Root → Right.

Result → 10 → 20 → 25 → 30 → 40 → 50 → 60 → 70 → 80

This is sorted, confirming correct BST structure.

(d) Is the given tree a valid BST.

Given.



This is NOT valid BST.

Node 65 is in the left subtree of 50.

But 65 > 50, which violates BST rule.

(e) Height comparison

Longest path:

50 → 30 → 20 → 10

Edge = 3

$\boxed{\text{Height} = 3}$.

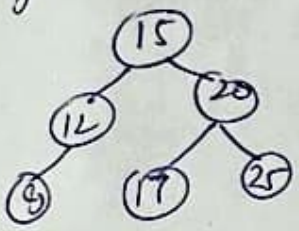
Problem 4 -

ⓐ Delete node 10

Step 1: Find inorder successor of 10
Right subtree of 10 → node 12
So successor = 12.

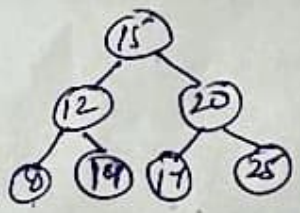
Step 2: Replace 10 with 12

Step 3: Delete original 12 node.



ⓑ Insert roll no. 14.

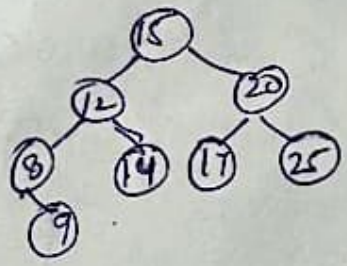
Steps
1. 14 < 15 → go left
2. 14 > 12 → go right
3. Insert as right child of 12.



ⓒ Insert roll no. 9

Steps:
1. 9 < 15 → left
2. 9 < 12 → left
3. 9 > 8 → right

Insert as right child of 8.



ⓓ Find roll no. b/w 10 and 20.

Best traversal: - Inorder traversal

8 → 9 → 12 → 14 → 15 → 17 → 20 → 25

Filter (10 to 20):

$\boxed{12, 14, 15, 17, 20}$

ⓒ Time complexity for searching roll no. 25

Best Case

$\Rightarrow O(1)$

Worst Case

$\Rightarrow O(n)$.

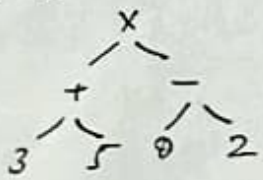
In this tree

Path

$15 \rightarrow 20 \rightarrow 25$

$O(\log n)$.

Problem 5 – $(3+5) \times (8-2)$



ⓐ Postorder Traversal (Left → Right → Root)

→ Left subtree (+) : 35 +

→ Right subtree (−) : 82 −

→ Root (×).

• 35 + 82 − ×

This is postfix notation.

ⓑ Inorder Traversal (Left → Root → Right)

$(3+5) \times (8-2)$

~~Match~~ Matches original expression.

ⓒ Preorder Traversal (Root → Left → Right)

× + 35 − 82

This is prefix notation

Operator comes before operands.

(d) Evaluation using Postorder (Stack Algorithm)

1. Initial empty stack
2. Scan postfix expression left → right.
3. If operand → push to stack.
4. If operator:
   → Pop two elements
   → Apply operation
   → push result back.

Ex → 3 5 + 8 2 - x

Final result = 48.

(e) Expression Tree for: a x b + c / d - e

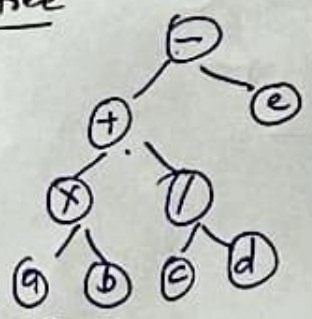
Step:

1: operator precedence.

- x and / first
- then +
- then -

$\boxed{((a \times b) + (c / d)) - e}$ expression.

2: Expression tree



3: Traversal

Inorder - ((a x b) + (c/d)) - e

Preorder - - + x a b / c d e

Post order - a b x c d / + e -